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Device including steering cables for creating a cavity or a channel in bone

Abstract

A device for creating a cavity or a channel in bone. First and second steering cables are coupled to a flexible elbow or a tip of a shaft. A control assembly is operably coupled to the first and second steering cables and configured to tension one of the steering cables and slacken another one of the steering cables so as to flex the flexible elbow. The control assembly may include a steering block axially movable within a handle with a knob threadably coupled to the steering block. The steering block may be rotatably fixed within the handle. The first and second steering cables may be anchored to the steering block, such as to opposite sides of the steering block. The first steering cable may extend along a linear path, and the second steering cable may extend along a curved rib of the handle.

Inventors: Brockman; Christopher Scott (Kalamazoo, MI), Vangemert; Geoffrey J. (Portage, MI)

Applicant: Stryker Corporation (Kalamazoo, MI)

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Assignee: Stryker Corporation (Portage, MI)

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Primary Examiner: Bates; David W

Attorney, Agent or Firm: Howard & Howard Attorneys PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of copending U.S. application Ser. No. 16/545,685, filed Aug. 20, 2019, which is a continuation of U.S. application Ser. No. 15/097,642, filed Apr. 13, 2016, now U.S. Pat. No. 10,441,295, which is a continuation of International Application No. PCT/US2013/065002, filed Oct. 15, 2013, each of which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to a device useful for creating a void space in living tissue such as bone. The device of this disclosure includes a handle with a control knob for regulating the setting of the device. The handle is constructed so that when the device is in different positions and positions and orientations relative to the practitioner using the device, the practitioner is able to easily hold the handle and set the knob.

BACKGROUND

(2) There are a number of medical devices used to create void spaces such as cavities or channels in living tissue such as bone. One such device is a cavity creator. A cavity creator is a surgical tool assembly that, as implied by its name, forms a cavity in the tissue of a living being. A number of cavity creators are specifically designed to form cavities in hard tissue, more specifically, bone. This type of cavity creator may be used in a procedure generally known as a vertebral augmentation procedure. In a vertebral augmentation procedure, cement is injected into a vertebral body that is suffering from fracture so as to stabilize the vertebral body. It is believed that this stabilization reduces the pain the fracture would otherwise induce in the patient. As part of this procedure, to ensure a sufficient quantity of cement is introduced into the vertebra, it may be necessary to form a cavity, sometimes referred to as a void space, in the vertebra.

(3) A cavity creator that forms a void space in a vertebra typically includes two components, an access cannula and a curette. The access cannula is a tube like device. The access cannula is inserted in the patient and into the vertebra to be filled with cement. The curette includes an elongated shaft dimensioned to extend through the lumen, the bore, of the access cannula. A tip is located at the distal end of the shaft. A handle is located at the proximal end of the shaft. Once the curette is seated in the access cannula the tip is positioned to extend radially, outwardly from the longitudinal axis of the access cannula. The rotation of the curette results in the pressing of the tip against the bone located outwardly from the access cannula. The pressing of the tip against the bone abrades the bone so as to form a cavity, the void space, internal to the vertebra that extends radially beyond the access cannula. Once this cavity is formed, the curette is withdrawn. Cement or other therapeutic agent is then introduced into the bone and more particularly the cavity that surrounds the open distal end of the access cannula.

(4) Some curettes include a tip that is pivotally attached to the distal end of the curette rod. A drive shaft internal to the rod is selectively extended/retracted to pivot the tip between an orientation in which the tip is axially aligned with the shaft and orientation in which the tip is directed away from the shaft. Some of these curettes rely on a thumb wheel and gear assembly mounted to the curette handle to selectively extend/retract the drive shaft. Still other of these curettes include ratchet mechanisms again mounted to the handle, to extend/retract the drive shaft.

(5) A disadvantage of a number of these curettes is that their internal components do not allow one to easily set the extended state position of their drive rods. This means that, in turn, it is difficult to set this type of curettes so as to with a degree of precision, control the extent to which the tip extends beyond the complementary access cannula. Further some of these curettes include numerous parts. As with any device, the more parts required to build the device increases the cost of providing the device.

(6) Still another known curette includes a shaft with a tip that is formed integrally with and extends distally from the shaft. At least the connector that holds the tip to the shaft is formed from a superelastic shape memory material. This curette is initially seated in the access cannula so that connector and the tip are wholly disposed in the cannula. When the cavity creator is in this state, the connector is bent to generally conform to the shape of the access cannula. The shaft of the curette is extended to cause the tip and connector to extend forward from the access cannula. As the connector emerges from the constraining space of the access cannula, the potential energy of the connector is released. This causes the connector to return to its unconstrained, bent, state. By extension, this results in the movement of the tip radially away from the access cannula. Once the tip is in the desired position, the curette is rotated so as to press the tip against the adjacent bone.

(7) The above-described curette includes a control knob for selectively extending/retracting the shaft. This knob, when rotated, moves towards and away from the curette. The practitioner then has to adjust his/her hand position relative to the handle in order to set the position of the tip. Having to take this extra step adds to the ergonomic complexity of using this type of curette.

(8) Moreover, as a consequence of the rotation of the knob to extend the above-described curette, the overall proximal-to-distal length of this device increases. Often a fluoroscope is in close

proximity to the cavity creator when this procedure is performed. The fluoroscope provides real time images of the bone in which the void space is being formed. The increasing length of the handle of this curette can make it difficult to position the curette around the fluoroscope.

(9) Further, many curettes, regardless of the type of assemblies that pivot their tips, share a common problem. When the cavity creator with which one of these curettes is integral is in certain positions, it is difficult for the practitioner to hold the handle so as to rotate the handle and/or set the extent to which the curette shaft is extended. For example, there are times when the practitioner forms a void space that is centered on an axis that extends into the patient through a portal located on the side of the patient opposite the side of the patient against which the practitioner is standing. This is known as the contra-lateral side of the patient. Thus, if the practitioner is standing adjacent the left side of the patient, it may be desirable to create a cavity that is located in bone best accessed from the right side of the patient. The presence of equipment, for example a fluoroscope, may make it difficult for the practitioner to simply reposition himself/herself adjacent the right side of the patient. In this situation, to form the cavity, the practitioner has to reach across the spine of the patient in order to insert the access cannula. When the curette is inserted into the access cannula, the handle does not simply face away from the practitioner as the handle would if located between the practitioner and the patient. Instead, the handle of the curette can be both spaced away from the practitioner and be directed towards the practitioner. To set the extended state position of the curette shaft or rod, the practitioner has to reach across the patient. This puts the hand of the practitioner in an awkward position relative to the component of the curette he/she needs to manipulate. This makes it difficult for the practitioner to set the curette to ensure that tip is properly positioned.

(10) Furthermore, it should be understood that the creation of a void space involves more than simply extending and retracting the shaft integral with the curette. It is also necessary to rotate the curette to press the tip against the tissue to be removed. This means that the practitioner may be required to place his/her hands and fingers in an unusual position in order to both set the extension of the curette and rotate the curette.

(11) Still another class of device used to form a void space in bone are channel creators. A channel creator is typically used to form an elongated bore, a channel, in the bone to which the channel creator is applied. These channels are typically smaller in cross sectional width and often longer in length than the cavities created by a cavity creator. The control members associated with a number of these devices have share a disadvantage with the cavity creators. It can be difficult to control the device depending on the orientation and position of the device relative to the practitioner.

SUMMARY

(12) This disclosure relates to a new and useful medical/surgical device. The medical/surgical device of this disclosure includes a handle designed to be held by the practitioner in a single hand. An elongated member extends forward from the handle. A working tip is located at the distal end of the elongated member. Internal to the handle is a mechanism for regulating the actuation of the working tip. A knob is rotatably mounted to the handle so the practitioner can regulate the actuation of the working tip. The knob is rotatably mounted to the handle so that regardless of the orientation and position of the handle relative to the practitioner using the device, the practitioner can, with the thumb or fingers of the hand holding the handle set the knob.

(13) In many versions of this disclosure the handle and knob are collectively dimensioned so that the knob is mounted to the handle so as to extend outwardly from opposed side surfaces of the handle and from a surface that bridges the side surfaces. This bridge surface may be the top surface, the front surface the bottom surface or the rear surface of the handle.

(14) One version of this disclosure is a new and useful cavity creator for creating cavity in the tissue, such as bone, of a living being. The cavity creator of this disclosure includes an access cannula and a curette. The curette is constructed so that, through a wide range of orientations and positions of the cavity creator relative to the practitioner, the practitioner with minimal

concentration and effort can manipulate the curette to set both the extent to which the tip extends from the access cannula and the orientation of the tip.

(15) The curette of this disclosure includes a handle dimensioned and shaped to be held in between the thumb and fingers of a single hand. A shaft extends forward from the handle. At the distal end, the shaft includes an elbow formed from a superelastic shape memory material. A tip is located at the free end of the elbow. A drive head is affixed to the proximal end of the shaft. The drive head is disposed in the handle so as to be able to move from essentially the proximal end of the handle to essentially the distal end of the handle.

(16) A knob is rotatably mounted to the handle. While the knob rotates, the longitudinal position of the knob relative to the handle is fixed. The handle and knob are collectively dimensioned so that the knob protrudes away from a top surface of the handle and adjacent sections of the opposed side surfaces. The knob is engaged with the drive head so that the rotation of the knob results in the extension or retraction of the drive head. The extension or retraction of the drive head results in the like movement of the curette shaft and attached components.

(17) The above features of the curette of the void creator of this disclosure make it relatively easy to, with a single hand and throughout a wide range of curette orientations relative to the practitioner holding the curette, both rotate the handle and rotate the knob. This makes it relatively easy to, when the cavity creator is in different orientations, both set the extent to which the tip is extended from the access cannula and the orientation of the tip relative to an axis through the access cannula.

(18) Another cavity creator of this disclosure includes an alternative curette. This curette includes an elbow that, while flexible, is not bent to a specific curvature. Steering cables extend proximally from the handle through the shaft and elbow to the tip. A mechanism internal to the handle selectively tensions and slacks the steering cables. The knob is the component the practitioner actuates so as to control the tensioning and slacking of the steering cables. The selective tensioning and slacking of the steering cables it is understood regulates the extent to which the elbow flexes the tip away from the longitudinal axis of the shaft.

(19) Another version of this disclosure is a channel creator. A prebent flexible elbow extends forward from the handle of this versions of the disclosure. A tip extends forward from the distal end of the elbow. The tip is formed from material that, when pushed forward, creates a channel in the tissue. A cable extends proximally from the tip, through the elbow and shaft into the handle. The knob is part of a drive assembly capable of selectively extending and retracting the cable. When the cable is in the retracted state, the cable places a tension on the elbow. The tension holds the elbow so that the elbow and tip are substantially in line with the longitudinal axis of the shaft. When the cable is in the extended state, the tension on the cable is reduced. Owing to the shape memory characteristics of the elbow, this allows the elbow to return the curved shape. When the elbow is so curved, the longitudinal axis of the tip is angled relative to the longitudinal axis of the shaft.

(20) The handle, knob and control mechanism assembly of this disclosure may have other applications than setting the angular position of a void creating tip relative to the longitudinal axis of the shaft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is pointed out with particularity in the claims. The above and further features and advantages of this disclosure are understood from the following Detailed Description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is an exploded view of the components forming a cavity creator of this disclosure;

(3) FIG. 2 is an exploded view of the components forming the curette of the cavity creator of this disclosure;

- (4) FIG. 3 is a view of the left side of the handle of the curette;
- (5) FIG. 4 is a view of the right side of the curette;
- (6) FIG. 5 is a plan view of the proximally directed surface, the back, of the handle;
- (7) FIG. 6 is a plan view of the distally directed surface, the front of the handle;
- (8) FIG. 7 is a plan view of the top of the handle of the handle;
- (9) FIG. 8 is a plan view of the bottom of the handle and knob;
- (10) FIG. 9 is a perspective view of the handle and knob of the curette;
- (11) FIG. 10 is a plan view of the interior of one of the left side shell of the handle of the curette;
- (12) FIG. 11 is an exploded view of the curette shaft and the components connected to the shaft;
- (13) FIG. 12 is a cross sectional view of the knob of the curette;
- (14) FIG. 13 is a perspective view of the knob;
- (15) FIG. 14 is a cross sectional view of the interior of the handle and the components internal to the handle of the curette;
- (16) FIG. 15 is a cross sectional view of the distal end of the cavity creator when the tip of the curette is in the retracted state;
- (17) FIG. 16 is a side view when the tip of the curette is in the extended state.
- (18) FIG. 17 is a cross sectional view of an alternative curette of this disclosure;
- (19) FIG. 18 is a enlarged cross sectional view of the shaft of the curette of FIG. 17;
- (20) FIG. 19 is a side plan view of a channel creator of this disclosure;
- (21) FIG. 20 is a cross-sectional view of the channel creator of FIG. 19;
- (22) FIG. 21 is a exploded view of the creator of FIG. 19; and
- (23) FIG. 22 is a perspective view of an electrosurgical tool constructed in accordance with this disclosure.

DETAILED DESCRIPTION

I. Cavity Creator with First Curette

(24) FIG. 1 depicts the components forming a cavity creator **30** of this disclosure. Specifically, cavity creator **30** includes an access cannula **34** and a curette **48**. The access cannula **34** includes a tube **36** formed from material able to penetrate both soft tissue and hard tissue (bone). The distal end of tube **36** may have beveled edge to facilitate the insertion of the tube into the patient. (Here, “distal” is understood to mean away from the practitioner holding the cavity creator **30**, towards the site to which the cavity creator is applied. “Proximal” is understood to mean towards the practitioner and away from the site to which the cavity creator **30** is applied.) A handle **38** is molded or otherwise secured over the proximal end of tube **36**. The handle **38** is formed with a fitting **40** that opens into the lumen internal to tube **36**. While not identified, the fitting is formed with threading. The threading facilitates the releasably coupling of either a cannula filled with cement or a syringe like device filled with cement to the access cannula.

(25) Curette **48** includes a handle **50**. A shaft **160** is moveably attached to the handle so as to be extended, move distally, and retracted, move proximally. At the distal end of the shaft **160** an elbow **164** is present. Elbow **164** is able to flex and return to its original bent shape. A tip **168** is located at the end of the elbow distal to the shaft. Shaft **160**, including elbow **164** and tip **168**, are moveably seated in the lumen of access cannula tube **36**. By adjusting the extension/retraction of the curette shaft **160** the extent to which the tip **164** extends radially outwardly access cannula tube **36** is selectively set.

(26) The curette handle **50**, as seen in FIG. 2, includes a left shell **52** and a right shell **54**. Shells **52** and **54**, when assembled together form the body of handle **50**. More particularly, the shells, as seen in FIGS. 3-10, form a base **56** and a stem **58** that projects distally forward from the base. Generally, the shells **52** and **54** are formed so that handle base **56** can be held between the thumb and fingers of a single hand. In some versions of the disclosure handle **50** is formed so that top to bottom, the base has a maximum length of between 6 and 13 cm. The top surface **57**, the proximal facing surface **61**, the bottom directed surface **63** and distal facing surface **65** of the base **56** are rounded.

Base **56** has its widest width in the plane where the distance across the base immediately below the top surface **57** is at its widest. This width is between 1.5 and 5 cm. Below this plane, the side surfaces **55** of the base **56** taper inwardly toward each other. The proximal to distal length of the base is greatest in the plane from which the opposed ends of the arch that forms the curved top surface extends. This length is between 2.5 and 6 cm. Further it should be understood that top surface **57**, proximal facing surface **61**, bottom directed surface **63** and distal facing surface **65** of the handle should generally be considered bridge surfaces in that each of these surfaces, extends between, bridges, the opposed side surfaces **55**.

(27) Stem **58** extends forward from the base **56** around an axis that is located in or near the plane from which the opposed ends of the top surface curve upwardly and inwardly towards each other. The stem has neck **70**. The neck has essentially parallel top and bottom surface **82** and **86**. Curved side surfaces **84** extend between the opposed ends of the top and bottom surfaces. Top and bottom surfaces are spaced apart a distance that allows adjacent fingers to be placed across the surfaces **82** and **86** with no discomfort. Thus, surfaces **82** and **86** are at a maximum, spaced no more than 1.5 cm apart from each other.

(28) Forward of neck **70** stem **58** is formed to have a head **96**. Head **96** is generally circular in shape. Handle **50** is formed so that the head **96** extends radially outwardly beyond the surfaces **82**, **84** and **86** of stem **58**. Shells **52** and **54** are further formed to define in front, distal panel of the head a semi-circular cut out **98** (identified in FIG. 2). When the handle is assembled cutouts **98** go into registration so as to define a distal end opening in the front of stem head **96**, opening not identified.

(29) Each shell **52** and **54** is formed with a notch **118**. Notches **118** extend downwardly from the top surfaces of shells **52** and **54**. In the depicted version of the disclosure the proximal end of each notch **118** is defined by a web **112** internal to the shell that extends longitudinally top to bottom through the shell. The web **112** of left side shell **52** seen in FIG. 10. Each web **112** is further formed to have semi-circular cutout **114**, one identified in FIG. 2, that opens into the notch. The distal end of each notch **118** is defined by a web **120** that, like web **120** extends inwardly from the outer portion of the shell **52** or **54** with which the web is integral. Web **120** is parallel to and spaced distally forward of web **112**. Each web **120** is formed with a semicircular cutout **122**, one identified in FIG. 2. Cutouts **114** and **122** are understood to be centered on the axis around which handle stem **58** is centered.

(30) Internal to the shells **52** and **54** are additional webs that extend inwardly from the inner surfaces of the side panels of the shells. Two webs, webs **126** and **128**, extend laterally, proximally to distally through the shell **52** or **54** with which the web is integral. The topmost of these webs, web **126** is the structural component of the shell against which webs **102** and **120** terminate. A web **130** that is parallel with and proximally spaced away from web **112** also extends upwardly from web **126**.

(31) Each shell **52** and **54** has two additional pairs of proximally-to-distally extending parallel webs. A first pair of webs, webs **134** and **136** extends forward from the inner surface of the rim of the shell that forms the proximal end of the shell. Webs **134** and **136** abut web **102**. Web **134** is located above web **136**. Webs **134** and **136** are equidistantly spaced from the axial line around which stem **58** is centered. A second pair of webs, webs **138** and **140**, extends forward from the distal face of web **120**. Webs **138** and **140** extend from the base **56** of the body of the handle and into stem head **96**. Web **138** is coplanar with web **134**. Web **140** is coplanar with web **136**. Within the stem neck **70** each web **138** and **140** has a break, not identified. Thus while not identified it should be understood that each web **138** has a proximal section located primarily in base **50** of the body and a distal section located in the stem head **96**.

(32) The left side shell, shell **52** is formed to have three posts **144** that extend inwardly from the inner face of the side panel of the shell. Each post **144** is formed with a closed end bore (not identified). Right side shell **54** is formed to have three openings **148**, identified in FIG. 2. Each opening **148** is formed in a recess **146** located inwardly from the side panel of the shell **54**. When

shells **52** and **54** are assembled together to form the body of handle **50**, each right side shell opening **148** is in registration with one of the posts **144** formed in the left shell **52**. Fasteners **152** that extend through the right shell openings **148** into the left shell posts **144** hold the shells together to form the body of the handle **50**.

(33) Curette shaft **160**, now described with reference to FIG. **11**, is generally tubular in shape. Shaft **160** is formed of material that, when the shaft is axially loaded resists axial buckling of the shaft. The shaft **160** is also formed from a biocompatible metal. Shaft **160** has a diameter that allows the shaft to slide in and out of the opening in the housing stem defined by cutouts **98**.

(34) A drive head **152** is located at the proximal end of shaft **160**. In cross section, in planes perpendicular to the longitudinal axis of shaft **160**, the drive head can be considered to be oval in shape. Drive head **152** is dimensioned to move within the space between webs **134** and **136**, through openings defined by cutouts **114** and **122** and in the space between webs **138** and **140**. The longitudinally extending opposed curved side surfaces of the drive head **152** are formed with threading **153**.

(35) The proximal end of curette shaft **160** is disposed in a bore **154** that opens in the distally directed face of the drive head **152** and extends proximally from that face. A set screw **155** holds shaft **160** to drive head **152**. Set screw **155** is disposed in a threaded longitudinally extending bore **156** that extends from bore **154** to a side surface of the drive head. Screw **155** seats in an indentation **161** formed in shaft **160**. Forward of drive head **152** indicia **163** (seen in FIGS. **2** and **11**) are formed on shaft **160**. The depicted indicia **163** are circular rings.

(36) Elbow **164** is formed from a shape memory material, sometimes called a superelastic material. This is a material that, once subjected to a force will deform and, upon release of the force, will return to the original shape. One biocompatible superelastic material from which elbow **164** is formed is a nickel titanium alloy known as Nitinol. In some versions of the disclosure, shaft **160**, elbow **164** and tip **168** are formed out of a single piece of Nitinol. The curette **48** is constructed so that elbow **164** extends outwardly from the plane that extends perpendicularly relative to the top-to-bottom longitudinal axis through handle **50**. The elbow **164** subtends an arc that, extending distally forward from shaft **160**, curves upwardly away from the shaft. In some versions of this disclosure, elbow **164** subtends an arc of between 20 and 50°.

(37) Tip **168** is integral with or attached to the free end, the distal end, of elbow **164**. In the illustrated version of the disclosure, tip **168** is oblate circle. Tip **168** has a diameter that is greater than the width of the elbow **164**. Here elbow “width” is understood to be the distance across the minor axis of the elbow **164**. Generally it should be understood that shaft **160**, elbow **164** and tip are formed so as that the elbow can be flexed sufficiently such that the each of these components can fit in and slidably move in the lumen of the access cannula tube **36**.

(38) A knob **170**, seen best in FIGS. **12** and **13**, is rotatably attached to the curette handle **50** to control the extension/retraction of shaft **160** relative to the handle. The knob **170** is formed to have tubular shaped hub **172**. Hub **172** defines a bore **174** of sufficient diameter that drive head can seat in and move within the hub. The hub **172** is further formed so that the distal section of the inner surface of the hub has threading **176** that extends into the bore **174**. Hub threading **176** is designed to engage threading **153** integral with drive head **152**. The outer diameter of hub **172** is such that the proximal end of the hub can seat in and rotate in the circular opening formed by shell cutouts **114** and the distal portion of the hub can seat in and rotate in the circular opening formed by shell cutouts **122**. The overall length of hub **172** is such that when curette **48** is assembled, the proximal end of the hub is approximately flush with the proximal surfaces of webs **112** and the distal portion of the hub is located forward of webs **120**.

(39) A flange **178**, generally in the shape of a washer extends radially outwardly from the outer surface of hub **172**. A ring **180** extends circumferentially around the outer perimeter of flange **178**. Ring **180** extends proximally and distally relative to the flange **178**. The ring has a proximal to distal length that allows the ring to be seated in the single notch in the body of the curette handle **50**.

formed by notches **118**. Ring **180** is formed with indentations **182** that function as finger grips. Webs **184** extend outwardly from the opposed surfaces of flange **178**. Each web **184** extends between the outer surface of the hub **172** and the inner surface of the ring **180** to provide reinforcing strength to the knob **170**.

(40) The components forming curette **48** are dimensioned so that when curette **48** is assembled, knob ring **180** projects outwardly from the opposed side surfaces **55** and the handle body and the interconnecting top surface **57**. More particularly, the ring indentations **182** are located outwardly from these body surfaces. In many versions of the disclosure, the portion of knob that protrudes radially outwardly out at least 1 mm from these surfaces of the body subtends an arc of at least 150°. In many versions of the disclosure the continuous exposed section of the knob extends through an arc of at least 180°. In more preferred versions of the disclosure the exposed portions of the knob subtend an arc of at least 200°. For ease of operation in many versions of the disclosure, curette **48** is designed so that the arcuate section of the knob that protrudes radially outwardly from the body of the handle at least 180° extends at least 2 mm outwardly from the adjacent top and side surfaces of the handle.

(41) The initial step of using cavity creator **30** of this disclosure is the insertion of the access cannula **34** into the bone in which the void space, the cavity, is to be created. Curette **48** is then fitted to the access cannula **34**. More particularly, the curette elbow **164** is bent so the shaft **160**, elbow **164** and tip can be fitted into the lumen, the bore, of access cannula tube **34**. When the cavity creator **30** is in this state, shaft **160** is in the fully retracted state so tip **168** is in closest position relative to the handle. When curette **48** is in this state, most of the body of the drive head **152** is disposed in the spaced internal to the handle **50** proximal to knob **170**, between webs **134** and **136**.

(42) Owing to the dimensioning of the components forming cavity creator **30**, when the cavity creator is in this state elbow **164** and tip **168** are, as seen in FIG. **15**, disposed within the distal section of access cannula tube **36**.

(43) The cavity creator process continues with the extension of shaft **160**. The practitioner extends the shaft by rotating knob **170**. Given that the position of the knob on handle **50** is fixed, the rotation of the knob is transferred through knob threading **176** and drive head threading **153** to the drive head **152**. The rotation of the knob against the drive head would be expected to cause the drive head **152** and, by extension, shaft **160** to move forward. However, the resistance of the bone adjacent the distal end of access cannula tube **36** prevents tip **168** from protruding outwardly. The ability of tip **168**, elbow **164** and shaft **160** to resist axial loading means that, instead of the tip moving distally, the rotation of the knob **170** results in the handle moving proximally. More specifically, the handle **50** moves proximally away from access cannula fitting **40**. More specifically the distally directed face of stem head **86** moves away from fitting **40**. This movement results in one or more of shaft indicia being exposed adjacent the access cannula fitting **40**. The number of exposed indicia provides the practitioner with an indication of the length of the exposed proximal section of shaft **160**.

(44) The practitioner then rotates the handle while simultaneously applying a distally directed force. These forces result in the tip **168** being pressed against the surrounding bone. This pressing of the tip against the bone abrades the bone so as to leave a void space that is located radially outwardly the distal open end of access cannula tube **36**. The practitioner presses inwardly on the handle until the distally directed face of handle stem head **96** again presses against the access cannula fitting **40**.

(45) Thus at the end of the process a void, a cavity, is formed immediately forward the distal end of the access cannula tube **36**. This void has a depth, axial distance from tube **36** to the distal end of the void, equal in distance to the distance to which the handle **50** originally was displaced proximally away from the cannula **34**. This is why the practitioner monitors the exposure of the shaft indicia **163**, to determine the extent to which the handle was originally displaced. As part of this procedure it should be further understood that the shaft may be rotated back and forth around

an arc of less than 360°. This results in a like rotation of elbow **164** and tip **168**. Consequently the void space formed by the cavity creator **30** of this disclosure will occupy an arc of less than 360° around the distal end of the access cannula tube **36**. Thus, using the cavity creator **30** of this disclosure, it is possible to both control the depth of the void space that is created as well as the degree to which the void space circumferentially surrounds access cannula tube **36**.

(46) In an alternative use of the cavity creator of this disclosure, the practitioner advances the shaft **160** while simultaneously holding curette stem **58** against the access cannula fitting **40**. The forward movement of the shaft **160** results in the movement of elbow **164** out of the distal end of the access cannula tube **36**. The freeing of the elbow **164** from the constraint of the tube releases the potential energy of the elbow; the elbow flexes towards the formed curved shape of the elbow. This flexing results in the movement of tip **168** radially away from the longitudinal axis of the access cannula as seen in FIG. **16**.

(47) At this time the tip **168** presses against the surrounding bone. Handle **50** is rotated so as to cause a like rotation of the tip **168**. Tip **168** presses against the bone. This pressing of the tip against the bone abrades the bone so as to leave a void space that is located radially outwardly the distal open end of access cannula tube **36**.

(48) Once the void space is created, the practitioner rotates the thumb so as to retract tip **168** back into the access cannula tube **34**. Curette **48** is then withdrawn from the access cannula **32**. The practitioner is then able to perform the next part of the procedure for which the creation of the void space was required.

(49) Curette handle **50** of the cavity creator **30** of this disclosure can be held in a single hand. In some procedures the cavity creator **30** is oriented so the distally directed face of handle stem **58** generally faces away from the practitioner. When curette **48** has this orientation relative to the practitioner, it is a simple task to, with fingers, hold the base **56**. When the handle **50** is so grasped, the stem neck easily sits between the forefinger and the middle finger. The practitioner uses the thumb of the hand holding the handle to rotate knob **170**.

(50) In some procedures, cavity creator **30** is oriented so that the distally directed face of handle stem **58** is generally directed towards the practitioner. When the handle is so oriented, the practitioner can easily hold the handle from the top such that the fingers are disposed against one of the side surfaces **55** and the thumb against the opposed side surface **55**. Knob **170** is rotated by the repositioning of the thumb from against the side surface **55** to against the knob. Thus, regardless of the orientation of the cavity creator **30**, the practitioner is able to, with minimal effort, grasp the curette handle **50** and rotate the knob **170** so as to both extend/retract the tip and rotate the tip.

(51) It should further be understood that when knob **170** is mounted to the handle **48** to rotate in a plane that is fixed relative to the handle. This means that as the knob rotates the practitioner does not have to reset the thumb or finger used to rotate the knob in order to compensate for the shifting position of the knob relative to the handle.

(52) In many versions of this disclosure, the drive head and knob threading **153** and **176** are formed so that a complete 360° rotation of the knob results in the linear displacement of the shaft by a maximum of 1.2 cm. This allows the practitioner to, by only slightly rotating the knob **170**, finely control the extent to which the tip **168** extends radially from the access cannula tube **36**. A related feature is that with a single movement, typically of the thumb the practitioner can rotate knob **170** at least 180°. This is because the knob protrudes from both the top surface **57** and side surfaces **55** of the handle. This means with a single movement of the thumb the practitioner can advance shaft **160** a distance of at least 0.6 cm. Collectively these design features mean that the cavity creator of this disclosure provides the practitioner with the ability to advance the tip by both relatively fine distances or, when required, by a relatively long distance in the short time required to engage in full stroke of the thumb or finger.

(53) Curette **48** of this disclosure essentially only has two moving parts; the shaft and attached components; and knob **170**. Thus given that this curette has relatively few components, in

comparison to curettes with more moving components, it can be more economical to provide.

II. Alternative Curette for Cavity Creator

(54) FIG. 17 depicts in cross section an alternative curette **210** that includes the features of this disclosure. Curette **210** is used with access cannula **34** (FIG. 1) to form an alternative cavity creator.

(55) Curette **210** includes a handle **212**. Handle **212** has the general outer shape and dimensions of handle **50** of curette **48**. Internal to the handle **212** are webs **214** and **216** that are analogues to webs **134** and **136**, respectively. Also internal to the handle are webs **218** and **220**. Webs **218** and **220** are analogues to webs **138** and **140**, respectively. Webs **218** and **220** do not extend completely into the stem of handle **212**. Internal to the handle a rib **224** extends inwardly from the bottom of the neck of the stem. Extending distally, the overall height of the rib decreases as the rib **224** curves downwardly. Handle **212** is also formed to have a curved rib **226**. Rib **226** is generally C-shaped. One end of rib **226** is located above and adjacent the distal end of web **216**. Rib **226** then curves downwardly and distally forward. From the lowest position with the handle rib **226** curves upwardly so as to approximately point to rib **224**.

(56) A shaft **260** extends forward the distally directed face of the stem of handle **212**. A hollow shaft **260** is formed from stainless or Nitinol and is tubular in shape. The proximal end of shaft **260** is seated in a block **230** of handle **212**. Block **230** is similar to below described block **302** of FIG. 21. Internal to handle **212** are webs with cutouts similar to cutouts **294** also seen in FIG. 21. Owing to the seating of the block **230** in the cutouts, block **230** is inhibited from rotating within the stem of handle **212**. Shaft **260** is statically mounted to the block **230**. Thus shaft **260**, like block **230**, is static relative to handle **212**.

(57) A flexible elbow **262** extends forward from the proximal end of shaft **260**. In some versions of the disclosure, shaft **260** and elbow **262** are formed as a single component. Slots **264**, one identified, are cut into the elbow to provide the elbow with flexibility. A tip **268** extends forward from the distal end of elbow **262**. Tip **268** is formed from material and is shaped so that when the tip is rotated and pressed against bone, the tip abrades the bone. Shaft **260**, elbow **262** and tip **268** are understood to be dimensioned so that these components can slidably fit within the axially extending lumen of access cannula tube **36**. A number of cavity creators of this version are constructed so that when shaft **260** is fully seated in the access cannula tube **36**, handle **212** abuts the access cannula fitting **40**, and the elbow **262** is flexed straight, tip **268** projects at least 1.5 cm forward of the tube **36**. In more preferred versions of the disclosure when the cavity creator of this disclosure is in this state, the tip **268** extends at least 2.5 cm forward from the access cannula tube **36**.

(58) Two steering cables **240** and **242** extend forward from handle **212** through shaft **260** and elbow **262**. As seen best in FIG. 18, cables **240** and **242** are located on opposed sides of the shaft **260** and, while not seen, opposed sides of the elbow. Also, while not seen the shaft **260** may be formed with opposed lumens wherein each cable extends through a separate one of the lumens. These lumens may be formed in ribs that project inwardly into the center bore of the shaft. Further, the shaft may be solid so that the lumens in which the cables **240** and **242** are disposed are two parallel bores that extend axially through the shaft. The distal ends of steering cables **240** and **242** are attached to opposed sides of the proximal portion of tip **268**. Alternatively, the opposed distal ends of the steering cables **240** and **242** attached to opposed sides of the distal end of elbow **262**.

(59) The proximal ends of the steering cables **240** and **242** are attached to a steering block **230** disposed in handle **212**. Steering block **230** has the same generally oval cross sectional shape of drive head **152**. The sides of the steering block **230** are formed with threading (not illustrated) similar to the threading **153** integral with drive head **152** (FIG. 11). Steering block **230** is formed with a closed end bore **232** that extends distally forward from the proximally directed face of the block. A closed end bore **234** extends proximally rearward from the distally directed face of the block **230**. Bores **232** and **234** are coaxial.

(60) The proximal end of steering cable **240** is anchored in steering block bore **232**. Cable **240** extends proximally from the steering block over the surface of rib **226** that faces proximally and downwardly. From rib **226** cable **240** extends over rib **224** and into shaft **260**. The proximal end of cable **242** is anchored in steering block bore **234**. Cable **242** extends approximately along a linear path distally forward from the steering block **230** into and through shaft **260**.

(61) Curette **210** includes a knob **227** similar to knob **170** (FIGS. **12** and **13**). A difference between the two knobs is that the hub **228** of knob **227** is slightly shorter than the hub **172** of knob **170**. Knob **227** is rotatably mounted to handle **212** in the same general manner in which knob **170** is mounted to handle **50**. The threading internal to knob hub **228** engages the complementary threading on the outer side surfaces of steering block **230**.

(62) A cavity creator that includes curette **210** is readied for use in the same general manner in which cavity creator **30** is readied for use. Prior to inserting shaft **260**, elbow **262** and tip **268** in the access cannula **34**, knob **227** is rotated to place the steering block **230** in the neutral position. The neutral position is the position in which owing to the position of the block, steering cables **240** and **242** impose essentially identical forces on the distal end of the elbow **262** or the tip **268**. When these forces are essentially identical, the elbow **262** and tip **268** are held so as to be essentially longitudinally aligned with the longitudinal axis of shaft **260**. When the curette is in this state, the shaft **260**, the elbow **262** and tip **268** can be inserted into the access cannula tube **36**.

(63) To create a cavity using this cavity creator, the cannula is pushed distally forward so as to cause tip **268** to project forward of the access cannula tube **36**. As part of this process knob may be rotated to set the orientation of the elbow **262** and tip **268** relative to the shaft **260**. For example if there is a desire to direct the tip upwardly as seen in FIG. **17**, knob is rotated to move steering block **230** proximally. This movement, in turn, causes the steering block **230** to pull steering cable **242** proximally while simultaneously reducing the tension the block places on steering cable **240**. This specific tensioning/slacking of the steering cables results in cable **242** pulling on the component to which the distal end of the cable **242** is attached so that the elbow **262** flexes as seen in FIG. **17**.

(64) Alternatively, curette **210** can be set so that elbow **262** and tip **268** are directed downwardly, opposite the orientation seen in FIG. **17**. This setting is achieved by rotating the knob **227** so that steering block **230** is displaced distally forward. This displacement of the steering block causes the block to simultaneously urge steering cable **240** proximally while reducing the tension that is placed on cable **242**. This results in forces being placed on the component to which the distal ends of the cables are applied that flex the cable downwardly such that the tip **268** points towards a location opposite the location to which the tip is directed in FIG. **17**.

(65) Once the tip **268** is oriented in the desired direction, the practitioner may oscillate the tip by rotating the handle **210** through an arc of less than 360°. This results in the formation of a void space forward of the distal end of the access cannula tube **36** that does not extend completely around the tube. The rotational position of this void space is understood to be set based on the setting of the angular position of curette tip **268**.

(66) Curette **210** of this version of the disclosure provides the practitioner with the means to set the angular orientation of the tip **268** relative to shaft **262**. This provides a further means to allow the practitioner to selectively form a void space in the bone regardless of the orientation in which the handle is being held. Thus, if the practitioner finds it necessary to hold the handle **212** upside down, wherein the top surface is directed towards the patient, by setting the angular orientation of the tip the practitioner can form a void space that is located upwardly relative to the portal in the bone through which the cavity creator is inserted in the patient.

(67) Still a further feature of this version of the disclosure is that by selectively adjusting both the extent to which tip **268** is angularly offset from shaft **260** and the extent to which the tip is advanced forward from access cannula tube **36**, the practitioner can form a void that while extending a 1 cm or more from the tube **36** does not project appreciably radially outwardly from

the tube.

III. Channel Creator

(68) A channel creator **280** incorporating the features of this disclosure is now described with reference to FIGS. **19** and **20**. Channel creator **280** includes a handle **282** from which a shaft **306** extends distally forward. A pre-bent elbow **310** formed from flexible shape memory material extends forward from the distal end of shaft **306**. A tip **314** extends from the distal end of elbow **310**.

(69) Handle **282** has the same general shape as previously described handles **50** and **212**. Internal to handle **282** are webs **286**, **288**, **290** and **292**. Webs **286** and **288** are substantially identical to previously described webs **134** and **136**, respectively. Webs **290** and **292** are similar to previously described webs **138** and **140**, respectively. Webs **290** and **292** are shaped to define rectangular cutouts **294** in the sections of the webs disposed within the head of the handle stem.

(70) Previously described knob **170** is rotatably mounted to handle **282**. A block **302** is seated in handle stem. More particularly block **302** is seated in cutouts **294**. Webs **290** and **292** and block **302** are collectively dimensioned so they abut. Webs **290** and **292** thus hold the block **302** static in handle **282**.

(71) Shaft **306** extends distally forward from the stem of handle **282**. Shaft **306** is formed from stainless steel or Nitinol and is tubular so as to have an axially extending lumen, (lumen not identified). The proximal end of shaft **306** is statically mounted in block **302**. Not identified is the bore in the block **302** through which shaft **306** extends. A bent elbow **310** is mounted to and extends distally forward from the distal end of shaft **306**. Elbow **310** is formed from a flexible shape memory material such as a nickel titanium alloy. In some versions of the disclosure, elbow **310**, when not flexed, subtends an arc of between 45 and 90°.

(72) In the depicted version of the disclosure, elbow **310** is formed so as to have a cylindrical stem **308**. While not identified it can be seen from FIG. **20** that the stem has a elongated groove that extends along the portion of the elbow that becomes the outer curved surface of the elbow.

(73) A tip **314** extends forward from distal end of elbow **310**. Tip **314** is formed from material such as Nitinol. In this version of the disclosure, the tip is shaped so that when the tip is pressed forward, the tip will penetrate, form a bore or channel in bone or other tissue. In the depicted version of the disclosure, tip **314** is formed integrally with the elbow **310**. The tip is formed to define a bore **316** that extends proximally from the distal end of the tip. Bore **316** is a closed end bore. While not seen in the drawings, the tip **314** is formed with a longitudinally extending notch. The notch is contiguous with and extends proximally from the proximal end of bore **316**.

(74) A number of channel creators of this version are constructed so that when shaft **306** is fully seated in the access cannula tube **36**, handle **282** abuts the access cannula fitting **40** and elbow **310** is flexed straight, tip **314** projects at least 2 cm forward of the tube **36**. In more preferred versions of the channel creator of this disclosure when the channel creator is in this state, the tip **314** extends at least 3 cm forward from the access cannula tube **36**.

(75) Drive head **152** is slidably disposed in handle **282**. More particularly, the drive head is slidably disposed between webs **286** and **290** located above the block and webs **288** and **292** located below the block. Knob **170** engages and moves the drive head **152** proximally and distally within the handle **282** as previously described.

(76) A tensioning cable **298** is attached to and extends distally forward of the drive head **152**. Cable **298** extends through the lumen of shaft **306**. By extension, the cable **298** extends through block **302**. The cable extends forward of the shaft and through the groove formed in elbow stem **308**. Cable **298** then extends around the outer curved surface of elbow **310**. At the distal end, the cable has a spherical head **299**. Cable head **299** is seated in the bore **316** formed in tip **314**. The portion of the elongated body of the cable **298** immediately proximal to the head **299** is disposed in the section of the notch formed in the tip **314** that extends proximally away from the bore **316**. Cable **298** and tip **314** are collectively formed so that cable head **299** cannot be pulled through the notch.

(77) Channel creator **280** is used in combination with access cannula **34** to form a channel in the bone to which the channel creator is applied. A channel differs from a void space formed by a curette in that a channel tends to be narrower in cross section and longer in length. The access cannula **34** is initially inserted into the bone in which the channel is to be formed. Knob **170** is then rotated so as to induce the proximal movement of the drive head **152** and, by extension, cable **298**. The rearward movement of the cable causes the cable to straighten out elbow **310**. The elbow **310** can be straightened, flexed, so that tip **314** becomes substantially aligned with the longitudinal axis of shaft **306**. Once the elbow is so straightened. Shaft **306**, elbow and tip can be slip fitted in the lumen of access cannula tube **36**.

(78) Tip **314** is shaped so that by reciprocating the channel creator **280** back and forth the tip forms a channel in the bone to which the tip is applied. The extent to which the path of the channel extends radially away from the distal end of the access cannula tube **36** is a function of the flexure of the elbow **310**. The flexure of the elbow is set by rotating knob **170**. More specifically, the knob **170** can be rotated to advance the drive head **152** and by extension, cable **298**, forward. This results in the lessening of the extent to which the cable **298** places a straightening tension on the elbow. The lessening of this tension cause the release of the potential energy of the material forming the elbow. This energy returns the elbow to the flexed, bent, shape. By controlling the extent which the elbow is allowed to return to the flexed shape, the practitioner regulates the extent to which the channel creator **280**, when advanced, forms a channel that extends radially from the access cannula tube **36**.

(79) Typically, the components forming cavity creator **280** are selected so that when elbow **310** is fully flexed, fully bent, the distal end of the tip **314** is radially spaced a minimum 1.5 cm from the longitudinal axis of shaft **306**. In many versions of the disclosure, the cavity creator **280** is constructed so that when the elbow **310** is so flexed, the distal end of the tip **314** is radially spaced at least 2.5 cm from the longitudinal axis of shaft **306**.

(80) The rotational position of the channel relative to the access cannula tube **36** is set by rotating the handle **282**. Thus, prior to rotating the knob **170** to set the curvature of the elbow **310** the practitioner may rotate the handle **282**. This rotation establishes the rotation direction in which the tip **316** is directed as the elbow returns to the curved state. The illustrated version is designed so that the practitioner knows that the elbow when fully flexed bends away from the bottom surface of the handle **282**.

IV. Alternative Embodiments

(81) The foregoing is directed to specific versions of this disclosure. For example, an alternative curette of this disclosure may not have the single component shaft-elbow-tip of the disclosed disclosure. In an alternative version of the disclosure, the tip may be pivotally attached to the shaft. In these versions of the disclosure, a drive rod with drive head is connected to the handle. The handle having the knob of this disclosure extends/retracts the drive rod so as to pivot the tip relative to the handle.

(82) In the described version of the disclosure, the knob is disposed in and rotates along an axis that is coaxial with shaft **160**. This is not meant to be limiting. In an alternative version of the disclosure, the knob is mounted to the handle so as to rotate around an axis that is orthogonal or otherwise angled relative to the axis of the shaft. Still in some versions of the disclosure, the knob may not rotate about an axis that intersects or otherwise overlaps the longitudinal axis of the shaft.

(83) Also there is no requirement that in all versions of the disclosure, the bridge surface from which the knob extends be the top surface. Thus in some versions of the handle bridge surface from which the knob extends may be one of the distally directed, bottom or proximally facing surfaces.

(84) Likewise, when the knob is provided, there is no requirement that it include the described indentations. In an alternative version of the disclosure the outer circumferential surface of the knob that is gripped by the thumb or finger may be formed from a material that facilitates gripping. These materials include textured plastic or rubber.

(85) In some versions of the disclosure, the rotating handle may be replaced by a lever or switch and a ratchet mechanism.

(86) Curette **48** of this disclosure may also be used as a stand alone device. Alternatively, the curette may be incorporated into other assemblies.

(87) While the foregoing description is directed specifically to the cavity creator of this disclosure, the features of this disclosure are not so limited. The handle **50** and knob **170** assembly of this disclosure may be incorporated into other medical devices where it is desirable to provide the practitioner with one-handed control of both the orientation of the device and setting of a moving component that extends from the handle. For example handle **50**, drive head **152** and knob **170** may be used to selectively extend/retract the shaft of another medical device. These devices include elongated forceps or scissors used in arthroscopic or endoscopic procedures. The component or components that extend from and the handle and controlled by the knob may not even be rigid. For example, the one or more flexible steering cables may extend from the handle. These distal ends of these steering cables extend to a flexible device such a fluid delivery catheter or an electrode assembly that is inserted in the patient. The selective tensioning of these steering cables is controlled by the selective rotation of the knob **170**. It should be appreciated that in these versions of the disclosure, the drive component/components that are connected to and actuated by the rotation of knob will be different from the disclosed drive head. For example, if the components controller by the knob are cables, the drive component/components may be one or more spools that are connected to the knob so as to rotate upon rotation of the knob.

(88) Likewise the component regulated by the knob may in some versions of the disclosure not be a component that performs a mechanical drive functions. The knob in some versions of this disclosure may be a component that controls the state of an electrical signal. Thus the knob function as the component that sets the position of the contact of a switch or the wiper of a potentiometer. Thus, as depicted in FIG. **22**, one construction of this disclosure is one in which attached to the distal end of a shaft **336** is an electrode **338** (monopolar or bipolar). In these versions of the disclosure, the rotation of knob **332**, by setting a switch and/or potentiometer internal to the handle sets the state of the control signal output from the handle **330** over a cable **298** back to a control console (not part of the present disclosure and not illustrated). Based on the state of the signal, the control console regulates the on/off state of the signal sourced to the electrode and/or the magnitude of the current or the level of the potential of the sourced current.

(89) Further, this disclosure is not limited to constructions of the disclosure wherein there is only a single manually actuated control member attached to the handle. Thus in a version of the disclosure wherein the device includes a shaft with an electrode tip, the tip may be selectively positionable through the use of steering cables or a tension cable relative to the shaft. In these versions of the disclosure, the knob be used to set the angular position of the electrode relative to the shaft. A second control member may be used to regulate the application of a current to the electrode.

(90) In some embodiments of these versions of the disclosure, the second control member may be a second knob. This second knob may or may not be parallel to the first knob. In still other embodiments of this disclosure, the second manually actuated control member may be another type of component such as a slide switch or a push button switch.

(91) In the described versions of the disclosure, the shafts that extend distally from the handle are all shown as being straight. This should be interpreted as a limiting feature of this disclosure. In some versions of the disclosure, the shafts, while rigid, may be formed with one or more curved sections or is formed to have a shape that is continuously curved. Likewise, while the shafts of the described versions of the disclosure may be interpreted to be rigid, this characteristic should not be interpreted as a limiting feature of this disclosure. In some versions of the disclosure, the shafts may be understood to be formed out of material that can be bent. This would allow the practitioner to, at the start of the procedure place custom bends in the shaft to facilitate the shaping of the device for the specific patient and procedure. In these versions of the disclosures it would be

understood that the driven components internal to the shaft would likewise be able to bend in order to properly function after this shaping.

(92) In still other versions of this disclosure, the distal end of the prebent, flexible elbow may actually be the tip of the device. For example, in one version of this embodiment of the disclosure, the device functions as a cement delivery cannula. This type of cannula holds cement for discharge into bone such as a vertebral body. A cement delivery cannula of this disclosure includes a shaft and a prebent, flexible elbow which have contiguous longitudinally extending through bores. In these versions of the disclosure, the proximal end of the shaft extends to a fitting that projects proximally from the handle. The fitting is designed for connection to a unit capable of supplying bone cement to the cannula. For reasons apparent below, this fitting is designed to rotate around an axis that extends from the handle. One cement delivery device to which this cannula can be attached is disclosed in the Applicant's U.S. Pat. No. 7,658,537, issued 9 Feb. 2010, the contents of which are explicitly incorporated herein by reference. The shaft itself is connected to the handle knob similar to how shaft **160** is connected to knob **170**. Thus the knob is rotated to selectively extend or retract the elbow relative to the handle. The distal end of the elbow is understood to be open.

(93) The practitioner uses the cement delivery cannula of this disclosure by placing this cannula in an access cannula similar to access cannula **34**. When the cement delivery cannula is placed in the access cannula the cement delivery cannula is initially set so the elbow is in the retracted state. The components of this system are designed so that when the elbow is so retracted, the elbow is fully disposed in the access cannula tube. This typically assumes the access cannula is already fitted in the patient. The unit capable of supplying cement is then connected by the handle fitting to the shaft of the cement delivery cannula. The knob is then rotated to set the extent to which the elbow extends forward from the distal end of the access cannula.

(94) Once the position of the cement delivery cannula is so set, the cement delivery unit is actuated. This typically involves the proximal movement of a plunger. This results in the forcing of cement out of the cement delivery unit and through the shaft and elbow of the cement delivery cannula and out of this elbow.

(95) Thus this version of this disclosure makes it possible for a practitioner to establish the location forward of the cement delivery cannula from which the cement will be discharged. This location setting in addition to being a radial distance is also a rotational setting. The practitioner establishes this setting by rotating the handle. This handle rotation it is understood results in a like rotation of shaft and elbow of the cement delivery cannula around the longitudinal axis of the access cannula.

(96) Accordingly, it is an object of the appended claims to cover all such variations and modifications that come within the true spirit scope of this disclosure.

Claims

1. A device for creating a cavity or a channel in bone, the device comprising: a handle; a shaft coupled to the handle and comprising a proximal portion seated within the handle, a tip located forward of the handle, and a flexible elbow between the proximal portion and the tip; a first steering cable coupled to the flexible elbow or the tip of the shaft; a second steering cable coupled to the flexible elbow or the tip of the shaft; and a control assembly comprising a knob that is rotatably coupled to the handle and operably coupled to the first steering cable and the second steering cable, wherein the knob is configured to be rotated to tension one of the first steering cable and the second steering cable and slacken the other one of the first steering cable and the second steering cable so as to flex the flexible elbow, wherein the control assembly comprises a steering block axially movable within the handle, wherein the knob is threadably coupled to the steering block, and wherein the first steering cable and the second steering cable are anchored to the steering block, and wherein the end of the first steering cable and the end of the second steering

cable are coaxially arranged.

2. The device of claim 1, wherein the steering block defines a distal closed end bore and a proximal closed end bore, wherein an end of the first steering cable is anchored to the steering block in the distal closed end bore and an end of the second steering cable is anchored to the steering block in the proximal closed end bore.
 3. The device of claim 1, wherein the handle comprises a curved rib, wherein the second steering cable extends along the curved rib.
 4. The device of claim 3, wherein the curved rib is C-shaped.
 5. The device of claim 1, wherein the handle comprises a base portion and a stem extending distally from the base portion, wherein the knob is rotatably coupled to the base portion and the shaft extends from the stem.
 6. The device of claim 5, wherein the stem comprises a neck comprising opposing surfaces spaced apart by a distance to accommodate adjacent fingers from a user, and a head extending radially outwardly from the neck.
 7. The device of claim 5, further comprising a mounting block seated within the stem, wherein the proximal portion of the shaft is mounted to the mounting block.
 8. The device of claim 1, wherein the knob is rotatable in a plane perpendicular to an axis of the proximal portion of the shaft.
 9. A device for creating a cavity or a channel in bone, the device comprising: a handle comprising opposing shells defining an interior of the handle, wherein at least one of the opposing shells is formed with a curved rib disposed within the interior of the handle; a shaft coupled to the handle and comprising a proximal portion seated within the handle, a tip located forward of the handle, and a flexible elbow between the proximal portion and the tip; a first steering cable coupled to the flexible elbow or the tip of the shaft; a second steering cable coupled to the flexible elbow or the tip of the shaft; and a control assembly operably coupled to the first steering cable and the second steering cable, wherein the second steering cable extends along a surface of the curved rib to orient an end of the second steering cable to be coupled to the control assembly to provide for simultaneous tensioning and slackening of the first steering cable and the second steering cable.
 10. The device of claim 9, wherein the curved rib is C-shaped.
 11. The device of claim 9, wherein the surface faces proximally and downwardly.
 12. The device of claim 9, wherein the first steering cable extends along a linear path.
 13. The device of claim 9, further comprising a control assembly comprising a steering block axially movable within the handle, wherein an end of the first steering cable and an end of the second steering cable are anchored to the steering block and coaxially arranged.
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